



Future of Work with AI Agents

Auditing Automation and Augmentation Potential
across the U.S. Workforce

{shaoyj, diyi}@stanford.edu



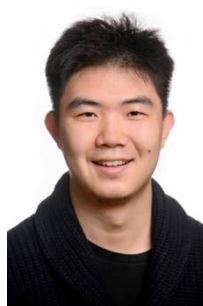
Yijia Shao



Humishka Zope



Yucheng Jiang



Jiaxin Pei



David Nguyen



Erik Brynjolfsson



Diyi Yang

IBM CEO: Layoffs Due to AI Led to 'More Investment' in Other Roles

BY PYMNTS | MAY 6, 2025



US Federal Workforce Layoffs: Can AI Agents Step In?

Craig Le Clair, Vice President, Principal Analyst MAY 5 2025

AI Replacing Jobs in 2025: Why Burnout Is the First Warning Sign of Mass Layoffs

AI tools are replacing jobs in recruiting, healthcare, and customer service. Here's how burnout, job anxiety, and workplace stress signal bigger shifts in 2025.

MIT report: 95% of generative AI pilots at companies are failing



BY **SHERYL ESTRADA**

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT

New Way of Curating Workforce Data

Human Workflow Auditing Framework

Automation

Augmentation



Semi-structured Survey



Mini-Interview

WORKBank

futureofwork.saltlab.stanford.edu

1,500 Workers

52 AI Experts

104 Occupations

Comprehensive audit of

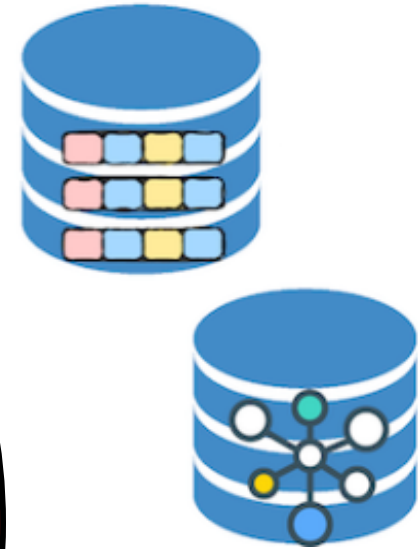
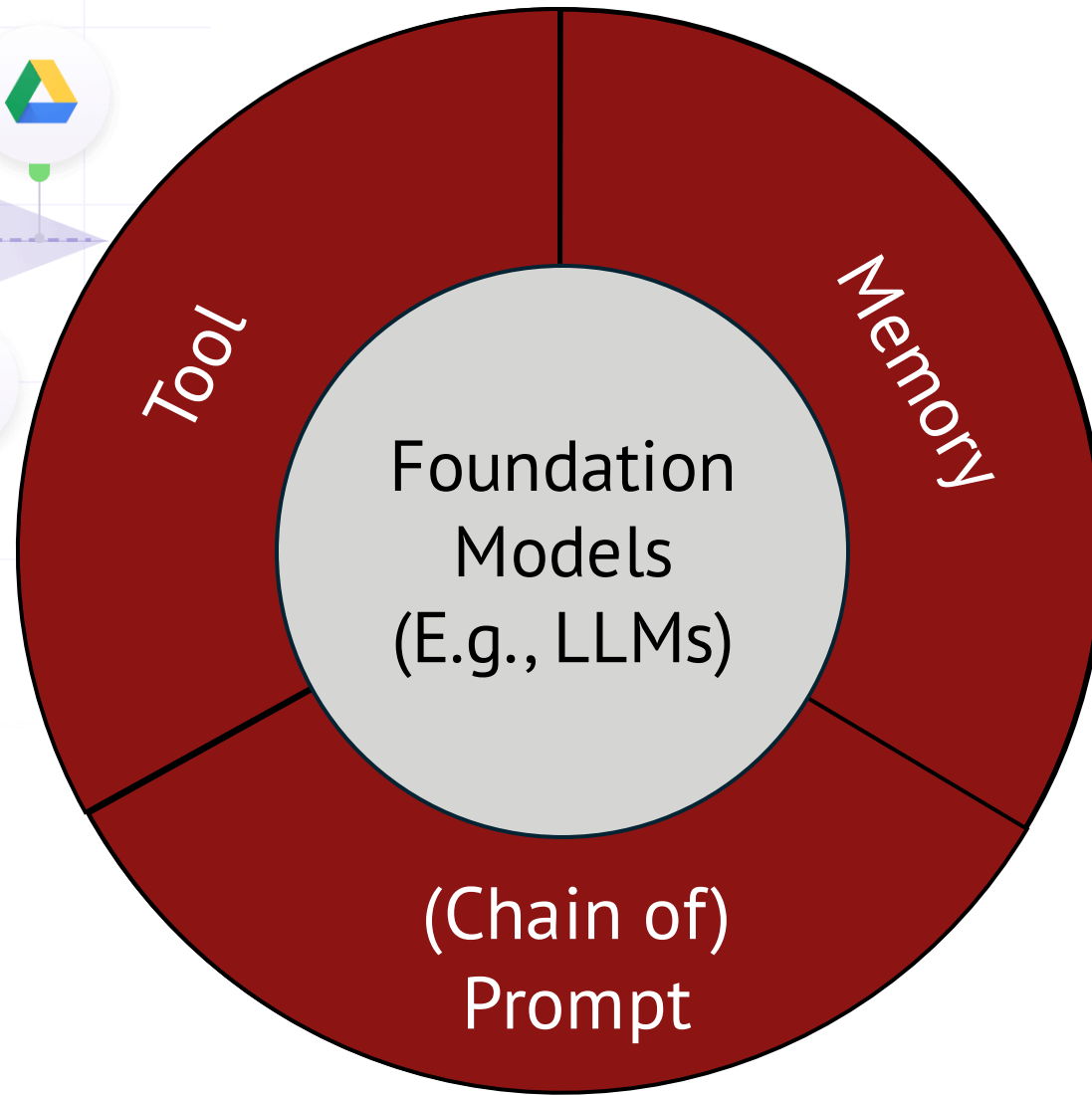
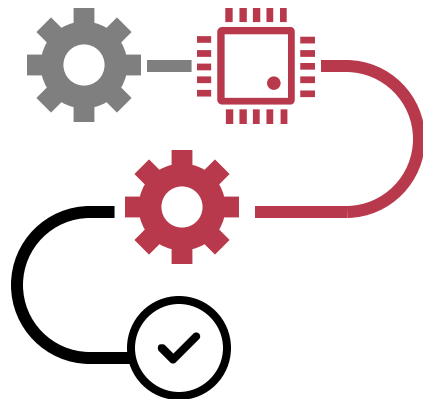
844 workflows from
O*NET developed by the
U.S. Department of Labor

Definition of Digital AI Agent

“

A system or program capable of autonomously performing tasks on behalf of a user or another system **by designing its workflow and utilizing available software tools**, without the ability to perform physical actions.

”



Agent

Current AI Agent R&D Concentrates on a Few Domains



SWE-agent



CLAUDE
CODE

Fork 1.6k



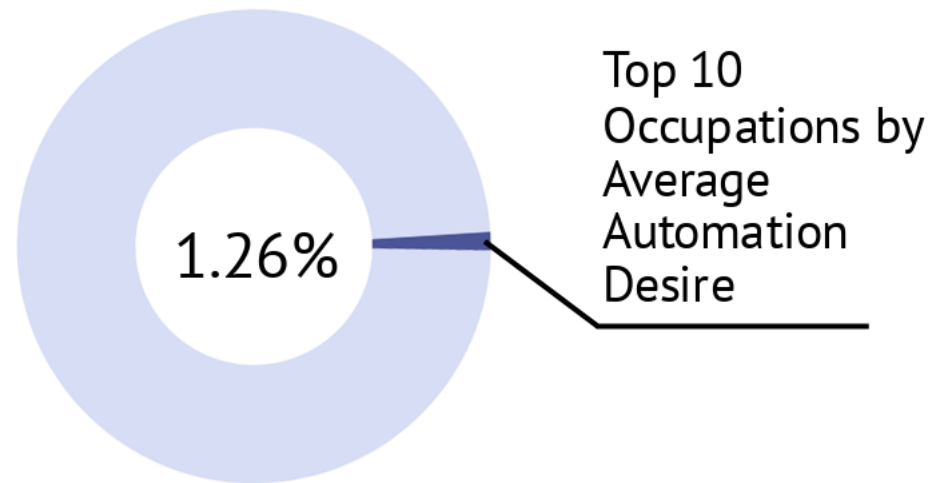
GitHub Copilot

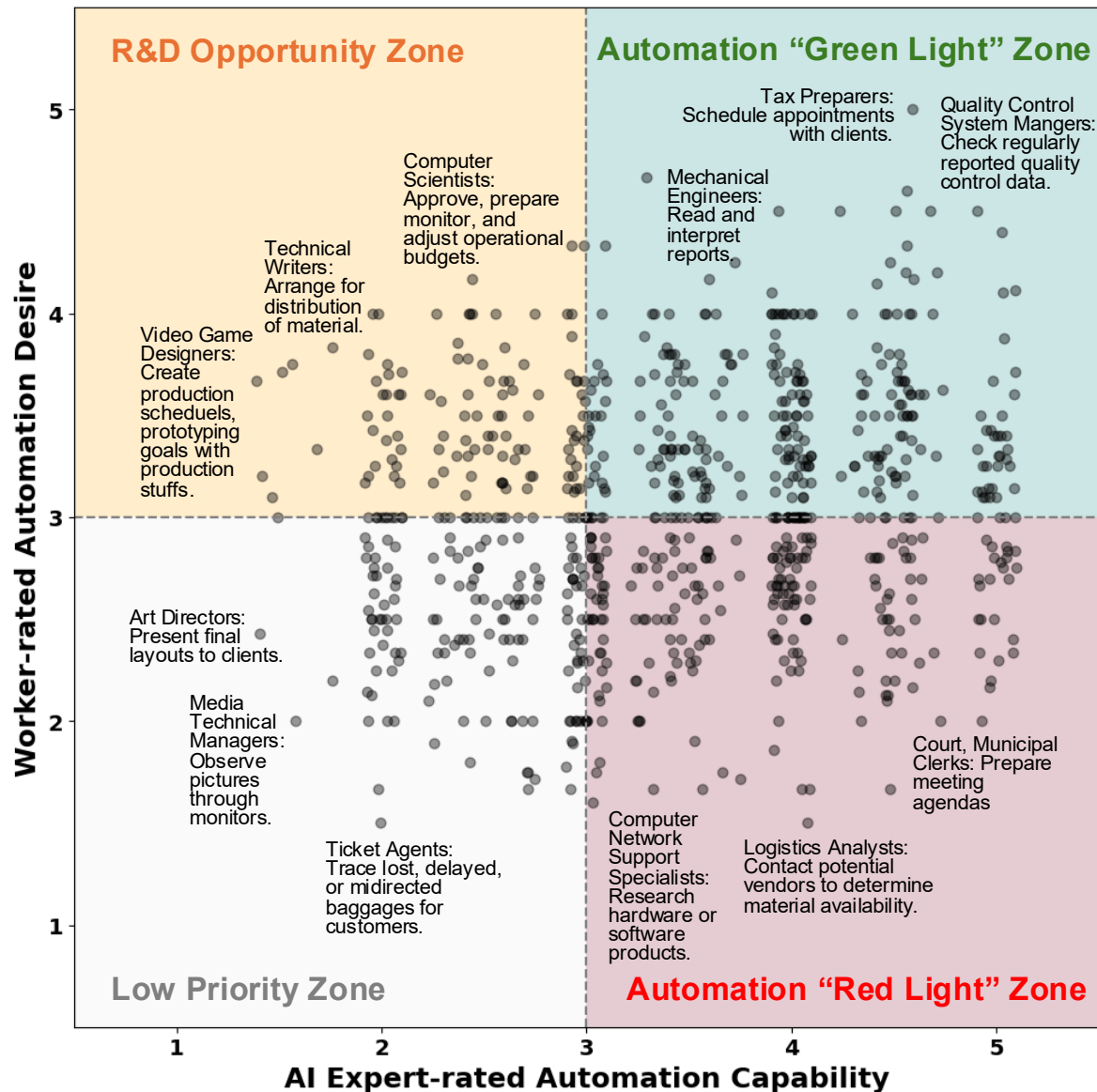


starred 24.2k



C Percentage of Usage on Claude.ai (Dec 2024–Jan 2025)



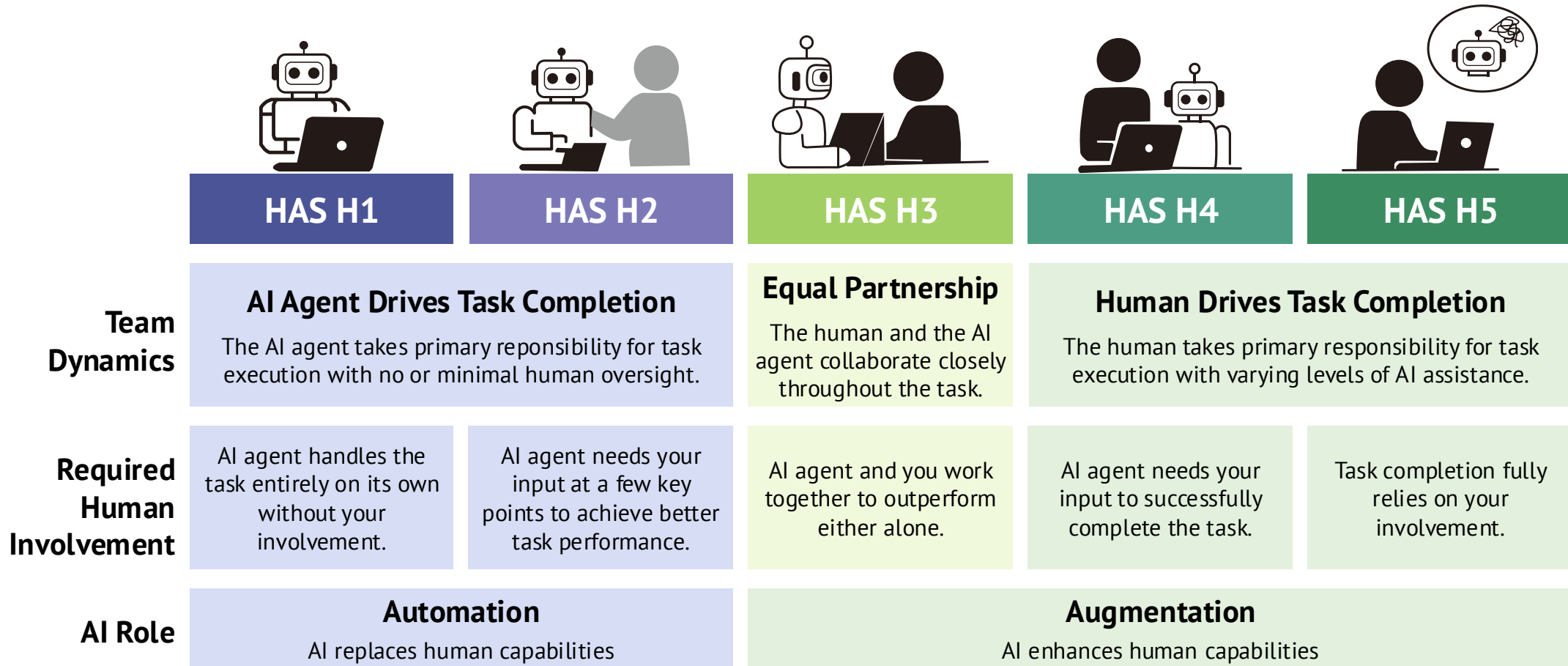


Bridge Worker Desire and Technology Capability

- **Automation “Green Light” Zone:** Tasks with both high desire and high capability.
- **Automation “Red Light” Zone:** Tasks with high capability but low desire.
- **R&D Opportunity Zone:** Tasks with high desire but currently low capability.
- **Low Priority Zone:** Tasks with low desire and low capability.

Human Agency Scale (HAS)

Quantifying Automation v.s Augmentation



HUMAN - AI COLLABORATION SPECTRUM

THE JOURNEY FROM AUTOMATION TO AUGMENTATION

HAS = HUMAN-AI COLLABORATION SPECTRUM



AUTOMATION ➤ AUGMENTATION



Occupation

Computer Programmers

Human Hands On or Off?

[View Human Agency Scale \(HAS\) Definition](#)



Detailed Task in the Occupation	Worker-desired Level	vs	Expert-assessed Feasible Level
Assign, coordinate, and review work and activities of programming personnel.	H3		
Compile and write documentation of program development and subsequent revisions, inserting comments in the coded instructions so others can understand the program.	H2		H1
Correct errors by making appropriate changes and rechecking the program to ensure that the desired results are produced.	H2		H4

Economic implications

Work is not only key source of income, but also purpose

Successful technology adoption requires:

1. Buy-in from workers and stakeholders to reduce resistance and frictions.
 - Our study highlights **Human Agency** and **Augmentation**.
 - Positive attitude for 46% of tasks → want more high value tasks.
2. Investment in complementary organizational change and product or process innovations (Bresnahan, Brynjolfsson, Hitt).
 - Our study urges to shift more AI investment from **red** to **green** zone.
 - 41% of investment in red zone!

Biggest barriers can be **institutional or political**. Workers are citizens and voters (cf. globalization backlash).

The Future of work and lump of labor fallacy

More studies at all levels: Effects heterogeneous. “AI Exposure” not specific enough. General equilibrium not clear.

- How are we going to reorganize tasks, jobs, firms?
- AI and workers complement or substitute? To some degree it's our choice.

The usual zero-sum thinking: Increasing supply leads to unemployment. But **there is no fixed pie!**

- 41% used to work in agriculture, now it's 2%.
- Related question of how we measure output, growth, and welfare.

The role of evolving **skills and education**

Ranked by Average Wage
(U.S. Bureau of Labor Statistics May 2024)

#1 Analyzing Data or Information

#8 Monitoring Processes, Materials

#11 Organizing, Planning & Prioritizing Work

#12 Communicating with People

#14 Documenting Information

#21 Training and Teaching Others

Ranked by Average Required Human Agency
(WORKBank AI Expert Assessments)

#1 Organizing, Planning & Prioritizing Work

#2 Training and Teaching Others

#8 Communicating with People

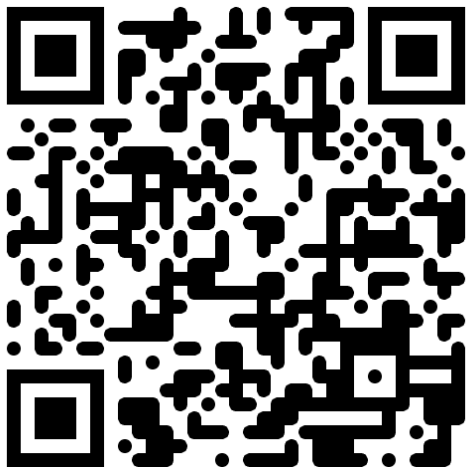
#13 Monitoring Processes, Materials

#17 Analyzing Data or Information

#19 Documenting Information

What's next?

1. Expand the audit to cover all workflows in our database.
2. Select representative tasks across the Human Agency Scale (HAS) and develop corresponding sandbox environments for worker training and longitudinal productivity measurement.



Feedback & Interest Form:
<https://forms.gle/hFGyhYkD1VwMLVj59>

Toward Human-Agent Collaboration



futureofwork.saltlab.stanford.edu



{shaoyj, diyyi}@stanford.edu